CausalCache: Conditional High-Fidelity Restoration for Long-Horizon GUI Agents
Supplementary Material

Anonymous submission

Scope of the Supplement

Arm naming in this supplement is explicit: HGKV+selector denotes the adapter-attached selector arm (three MobileWorld rounds); the deployed CausalCache of the main text is the frozen policy + selector (single round). Recent/summary arms are shared.

The main paper is self-contained. This supplementary document provides the complete fixed-budget design, confidence intervals for the policy and selector ablations, per-round closed-loop results, and deployment-cost measurements. The supplement is a separate PDF and does not count toward the seven pages of main content. All results use the same summary trace and archived screenshots as the main paper. Restoring an event means retrieving its archived post-action screenshot and reattaching it to the event summary; no screenshot is generated from text.

Fixed-Budget Restoration Protocol

At decision t, every completed event remains visible in text. An active visual-context budget B allows exactly B events to occupy the official retained-turn form: the step's verbatim policy response paired with its reattached archived screenshot. Every other event appears as a one-line action summary. Recent-B assigns the retained turns to the latest B eligible events. A restoration allocation may instead demote a recent event to its one-line summary and promote a distant event to its retained turn, preserving the image count, image resolution, retained-turn count, and prompt structure; the official protocol pairs each retained observation with the response that produced it, so the two forms move as a unit. The realized number of non-recent promotions is

$$k = |S \setminus \text{Recent-}B|.$$

It is an observed selector output, not a preset experimental treatment.

B	Recent arm	Replacement training arm	Deployed allocation
1	Recent-1	recent → archived	exact-1
2	Recent-2	oldest → archived	exact-2 (recent fallback)
4	Recent-4	oldest → archived	exact-4 (recent fallback)
8	Recent-8	built, not trained	exact-8 (recent fallback)

Table S1: Matched-budget fidelity-restoration design. The B=8 row is a budget-extrapolation test; it is excluded from adapter and selector training.

Intervention arms. For a matched group, R_0 and R_A use Recent-B under the frozen and adapted policies; S_0 and S_A replace the oldest recent image with a target-relevant archived image; W_A uses an age-matched wrong archived image. The adapter selection statistic is

$$\text{DiD}_{\text{select}} = (S_A - S_0) - (R_A - R_0).$$

The gate requires a positive episode-clustered bootstrap lower bound and point-estimate drift caps $|R_A - R_0| < 0.02$ and $|W_A - W_0| < 0.02$. Reported coalition utilities always rerun the complete restored set; they are never sums of singleton utilities.

Single-slot training interface (k=1)**.** Under the strictest *image-only* criterion—the required value appears in neither the goal, the current frame, the recent window, nor any completed step's action text—mining 1,326 GUI-Odyssey trajectories yields 96 image-only decision points at B=4, of which only two require two distinct evidence frames simultaneously (3 of 239 at B=1); on the desktop corpus, 75% of eligible states admit exactly one recurrence witness. A matched-budget probe under the official multi-turn protocol shows the first replacement is worth its slot (+0.013, 95% CI $[+0.005, +0.021]$) while the paired increments of a second and third replacement are indistinguishable from zero or negative (+0.001 and −0.011, n.s.). Supervision is annotation-free, so a domain exhibiting multi-frame demand requires only regenerating the corpus with $k \geq 2$ restoration arms: a data-mix change, not a method change.

Uncertainty. Offline intervals use 2,000 episode-clustered bootstrap replicates. Closed-loop intervals resample tasks while preserving the paired arms and repeated-run structure. A confidence interval crossing zero is described as indistinguishable, not as evidence for equivalence.

Additional Desktop Results

Policy-Side Gates Across Visual Budgets

Adapter	B	$\mathrm{DiD}_{\mathrm{select}}$ [95% CI]	$\lvert A_r \rvert$	Wrong	Gate
HGKV	1	$+.0224$ $[+.0162, +.0294]$	.0088	.0085	pass
HGKV	2	$+.0164$ $[+.0110, +.0221]$	.0107	.0078	pass
HGKV	4	$+.0109$ $[+.0070, +.0154]$	.0098	.0091	pass
HGKV	8	$+.0063$ $[+.0018, +.0115]$	.0106	.0137	pass
Ungated	1	$+.0177$ $[+.0135, +.0220]$	.0086	.0071	pass
Ungated	2	$+.0139$ $[+.0102, +.0177]$	.0079	.0069	pass
Ungated	4	$+.0075$ $[+.0031, +.0118]$	.0098	.0089	pass
Ungated	8	$+.0047$ $[+.0007, +.0093]$	.0160	.0163	pass

Table S3: Complete per-budget adapter gates. Group counts for $B = 1/2/4/8$ are $94/72/45/11$; the frozen selection effects are $+.0791/+.0256/-.0002/-.0305$. B=8 is untrained.

Both adapters pass, but HGKV has a larger selection effect at every budget and remains farther from the drift caps at B=8. Across the trained budgets $B \in \{1, 2, 4\}$, HGKV maintains positive selectivity and the learned allocation improves over Recent-B at the deployment-relevant budgets; at the unseen B=8, policy-side selectivity transfers, whereas the end-to-end offline allocation gain is inconclusive.

Why the recent reference matters. (The probes in this paragraph were measured during the initial development iteration of the corpus and adapter; the frozen-policy contrasts are checkpoint-independent, and the adapter re-scoring value is from that iteration's selected checkpoint.) At B=1, the deployment renderer's most recent slot can duplicate the current screenshot. Against that redundant control, the frozen policy prefers the relevant archived event by $+0.0333$ nats/token (95% CI $[+0.0118, +0.0567]$). Against the informative true previous frame, the frozen preference is only $+0.0013$ $[-0.0242, +0.0253]$. The HGKV adapter retains a positive true-previous-frame DiD of $+0.0095$ $[+0.0056, +0.0138]$. Thus the learned selectivity is not an artifact of comparing useful pixels with a duplicated current screen.

Selector and Feature Ablations

Allocation rule	B=2	B=4
Learned scorer	$+.0297$ $[+.0185, +.0409]$	$+.0169$ $[+.0067, +.0277]$
Random exact-B	$+.0058$ $[-.0056, +.0168]$	$+.0082$ $[-.0009, +.0177]$
RGB similarity	$+.0061$ $[-.0032, +.0150]$	$+.0071$ $[-.0010, +.0152]$
Inverted score	$-.0159$ $[-.0290, -.0033]$	$-.0091$ $[-.0180, -.0004]$
No recency features	$+.0121$ $[-.0003, +.0246]$	$+.0084$ $[-.0012, +.0173]$

Table S4: Exact-B selector ablations with fresh full-set policy scoring. Random and RGB-similarity rules do not explain the gain; reversing the learned ranking is harmful.

At B=4, the learned scorer chooses $k = 0/1/2/3/4$ on $103/103/88/119/146$ evaluated states. It therefore retains the entire recent window on 19% of states and performs non-recent restoration selectively. The corresponding distributions for random allocation are $96/138/161/152/48$, and for RGB similarity $233/166/106/66/24$. Performance cannot be attributed to using a fixed sparse history pattern or to always displacing recent images.

Selector input features. The scorer's input encodes, explicitly: candidate age and recency-window membership, target-recurrence witness signals with coordinate distance, action family, frame-deduplication multiplicity, trajectory shape, and set context (selected-set size, overlap and age gaps between the candidate and the already-selected events). The witness features compare the action taken right after an archived event against the reference action.

Single-pass reference variant. Replacing the proposal reference with the previously executed action removes the second policy call, adding only the selector's own cost ($<$ 10 ms). Offline it matches the proposal reference ($+.039/+.025/+.015/-.005$ over Recent-B at B=1/2/4/8) and on OSWorld-Verified it reaches 34.1%; zero-shot on mobile it drops to 29.1% against 33.9% for the deployed proposal-conditioned method, with the failure concentrated on single-app control tasks where the last-action reference trivially recurs. Cross-platform transfer is what the proposal reference buys.

Witness-reference ladder. The single-pass last-action reference yields $+0.0247$ $[+0.0133, +0.0363]$ at B=2 and $+0.0152$ $[+0.0058, +0.0250]$ at B=4 on desktop. Removing the witness reduces these effects to $+0.0161$ $[+0.0025, +0.0299]$ and $+0.0086$ $[-0.0024, +0.0196]$, respectively. This motivates the proposal-conditioned two-pass path as the default for cross-platform transfer, while retaining last-action conditioning as a domain-matched efficiency variant.

Additional Closed-Loop Results

Zero-Shot Mobile Evaluation

Memory-critical split construction. The split is fixed by a mechanical rule over the benchmark's own task definitions: `memory_candidate` = tasks whose official metadata lists more than one application (`app_count` $\geq$ 2 from the upstream `BaseTask.app_names`), `single_app_control` = exactly one, restricted to the GUI-only interface. Every record in the frozen manifest carries the upstream source path and content hash; no model outcome and no manual labeling enter the rule. Cross-application membership is a preregistered memory-challenge *candidate*, not a per-task guarantee of memory dependence: candidates that do not in fact require distant evidence dilute the stratum toward zero, so the reported contrast is conservative.

Method	Run 1	Run 2	Run 3	Mean
Frozen, summary-only	33/117	32/117	34/117	28.2%
HGKV + Recent-4	34/117	37/117	35/117	30.2%
HGKV+selector	39/117	42/117	38/117	33.9%

Fixed-budget fidelity allocation (offline desktop heldout; primary setting $B = 4$)							
Policy	Allocation	B=1	B=2	B=4	B=8[†]	Realized k	Notes
Frozen	Recent-B	0	0	0	0	0	reference
Frozen	oldest slot → archived	$+.079$	$+.026$	$-.000$	$-.031$	1	
HGKV	Recent-B	$+.002$	$+.005$	$+.007$	$+.009$	0	drift-capped
HGKV	oldest slot → archived	$+.103$	$+.047$	$+.018$	$-.016$	1	gate: main Tab. 3
HGKV	selector, last-action reference	$+.039$	$+.025$	$+.015$	$-.005$	measured	71% displace $\geq$1 at B=4
HGKV	selector, oracle-reference variant	$+.044$	$+.025$	$+.011$	$-.002$	measured	teacher-forced oracle

Table S2: Fixed-budget fidelity-allocation results. Each cell is a matched-count contrast on the desktop heldout split; every allocation promotes exactly B events to summary-plus-image form, and the learned selector may keep every promotion on the recent window (k=0); [†] B=8 is excluded from training and reports budget extrapolation. Cross-B comparisons are analysis only.

Table S5: Per-round zero-shot MobileWorld success. The paired HGKV+selector minus HGKV+Recent-4 mean is $+3.7$pp (95% CI $[+0.8, +10.3]$).

Stratum	HGKV+selector	Recent-4	Δ
Memory-critical	28.0%	19.4%	$+8.6$pp $[+1.6, +16.1]$
Control	40.6%	42.4%	-1.8pp $[-6.7, +3.6]$

Table S6: Primary mobile allocation analysis ($n = 62$ memory-critical; $n = 55$ control). The split-by-method interaction is $+10.4$pp (95% CI $[+1.5, +19.6]$, two-sided $p = 0.022$).

Outcome-defined sensitivity analysis. On the 95 tasks where the frozen summary-only policy reaches the official step cap, HGKV+selector scores 32.6%, Recent-4 28.9%, and summary-only 23.9%. The paired HGKV+selector minus summary-only difference is $+8.8$pp $[+1.8, +16.0]$; the paired difference from Recent-4 is $+3.7$pp $[-2.6, +10.0]$. On the 22 tasks where summary-only terminates early, HGKV+selector is 43.2%, Recent-4 is 36.4%, and summary-only is 47.0%; HGKV+selector minus summary-only is -3.8pp $[-20.5, +12.1]$. These outcome-defined strata are secondary because they condition on baseline behavior; the construction-defined split in Table S6 is the primary analysis.

Two-pass mobile cost. Across 2,224 audited steps, the proposal-conditioned second policy pass fires on 81.7% of all steps and 96.9% of steps with more than B eligible events. Median pass-1, selector, and conditional pass-2 times are 4.229 s, 0.039 s, and 3.258 s, respectively. Mean per-step overhead relative to pass 1 is 65.9%. The selector itself is cheap; the extra policy pass dominates.

OSWorld-Verified Evaluation

Method	Rate	Paired comparison
B0	19.9% (72)	reference
FR	32.7% (118)	$+12.7$pp $[+8.6, +17.2]$ vs. B0
HR	33.0% (119)	$+0.3$pp $[-3.1, +3.6]$ vs. FR
HGKV+selector	32.7% (118)	-0.3pp $[-3.9, +3.3]$ vs. HR

Table S7: OSWorld-Verified closed-loop results on the official no-GDrive 361-task roster with a 15-step budget; parenthesized values are success counts. High-fidelity history is beneficial relative to summary-only memory; same-budget allocations are statistically indistinguishable. B0 denotes frozen summary-only, FR frozen Recent-4, and HR HGKV Recent-4.

For HGKV+selector on OSWorld, the median task incurs eight extra policy calls and 20.6 model-seconds. Median wall time is 307.6 s, so the added model time is approximately 7% of end-to-end task time. These results separate two claims: desktop closed loop supports the availability of high-fidelity history, while the mobile construction-defined split tests which summarized events should receive that fidelity.

Closed-Loop Budget Sweep

Single-round MobileWorld campaigns evaluate HGKV+selector at $B \in \{1, 2, 8\}$ with the same frozen weights; only the selection budget and the runner's matched image budget change. The same-budget references are the frozen Recent-B dose curve at $B \in \{1, 2\}$ (licensed by the adapter's behavioral neutrality on matched allocations, verified on both platforms) and a dedicated HGKV+Recent-8 campaign at B=8, which also extends the extrapolation test from the offline gates to the closed loop. B=4 remains the primary setting; single-round budget cells are trend evidence, not headline contrasts.

	Full roster (117)		Mem.-critical (62)	
B	Recent-B	HGKV+sel	Recent-B	HGKV+sel
0	28.2	28.2	21.0	21.0
1	26.5*	29.1	16.1*	21.0
2	26.5*	33.3	17.7*	25.8
4 (primary)	30.2	**33.9**	19.4	**28.0**
8 (untrained)	29.9	33.3	27.4	29.0

Table S8: Closed-loop MobileWorld success (%) along the budget axis. At B=0 HGKV+selector is structurally identical to the summary-only baseline. B=0/4: three-run means; other budgets: single rounds. *Frozen-policy recent arm.

Full-roster success (/117): HGKV+selector 34/39/(39,42,38)/39 vs. same-budget recent 31/31/(34,37,35)/35 at B=1/2/4/8, i.e. per-budget paired gains of $+2.6/+6.8/+3.7/+3.4$pp. Pooled with equal budget weight over $B \in \{1, 2, 4, 8\}$ (task-level paired bootstrap, 20,000 replicates, independent resampling within each budget), the allocation gain is +4.1pp, 95% CI $[+1.3, +7.1]$, two-sided p=0.0039. On the construction-defined memory-critical subset the HGKV+selector−recent margin is $+4.9/+8.1/+8.6/+1.6$pp: largest at the trained budgets, narrowing at the untrained B=8 where the eight-frame recent window itself reaches part of the split's evidence.

Per-budget deployment cost. Across audited steps (5,465/5,111/2,224 at B=1/2/4), the conditional second pass fires on 86.7/81.0/81.7% of steps; median pass-1 latency is 3.7/3.9/4.2 s, the selector costs under 40 ms, and the conditional second pass adds a mean of 2.0/2.3/2.9 s per step, i.e. 8.1/8.7/10.5% of the measured per-step wall time (screenshot mtime spans). The B=8 campaign ran on a server revision without step instrumentation; an instrumented rerun on a roster subset (97 audited steps) shows the second pass firing on only 39% of steps—with eight recent slots the selector frequently keeps the recent tail—so the mean added time falls to 1.8 s per step, roughly 5% of the B=8 per-step wall. Real-time overhead therefore peaks near 10% at the primary budget and decreases at larger budgets as displacement becomes rarer.

Restoration-Content Controls

Two single-round campaigns at the primary budget keep the trained selector, the exact-B interface, and the conditional second pass unchanged, and vary only what the restored slots contain. The *text-only* control removes every history image from the composed prompt and substitutes the verbatim recorded response of the restored turn; the *OCR* control substitutes a Tesseract transcription of each restored screenshot. Selector features are computed on unmodified data, so the restoration *decisions* are distributed as in the pixel arm; only the policy-visible carrier changes. Because restoration moves an official retained turn (verbatim response plus screenshot) as a unit, these controls also settle the attribution question: states recoverable *only* from pixels are rare (0.36% of audited B=4 steps), yet removing pixels removes the gain.

Restored content	Full (117)	Mem.-crit. (62)	Control (55)
Pixels (HGKV+selector)	**33.9**	**28.0**	40.6
OCR text	32.5	24.2	41.8
Verbatim response	27.6	18.0	38.2
None (Recent-4)	30.2	19.4	42.4

Table S9: Success (%) when the restored slots carry pixels, OCR text, or the verbatim response, versus no restoration. Pixel and Recent-4 rows: three-run means; controls: single rounds. The text-only round scored 116/117 tasks: one task's benchmark evaluator returned persistent server errors after the trajectory completed (an evaluation infrastructure fault; paired contrasts drop the task).

Task-paired bootstrap contrasts (20,000 replicates): on the memory-critical split, verbatim text sits −10.4pp below pixels (95% CI $[-18.6, -3.3]$, p=0.006) and −1.6pp from Recent-4 ($[-8.7, +5.5]$)—replacing pixels with the turn's own text erases the subset gain entirely. OCR text recovers most of the level (−3.8pp from pixels, $[-12.9, +5.4]$; +4.8pp over Recent-4, $[-2.2, +12.4]$), and the direct OCR−text-only margin is +4.9pp ($[-3.3, +13.1]$); at single-round power neither separates from zero. The ordering that survives significance testing is: pixel restoration is the only carrier whose memory-critical margin over the recency heuristic is significant, and verbatim text is significantly below pixels. The restored screenshot's value is therefore not reducible to "the turn was mentioned again," and is only partially captured by its extractable text.

HGKV Token-Level Structure

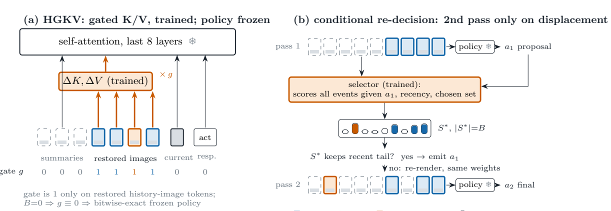

Figure S1: The token-gated K/V probe and the proposal-conditioned re-decision, in full detail: a low-rank $\Delta K, \Delta V$ branch on the last eight frozen layers is gated to restored history-image tokens only and bitwise bypassed at B=0; deployment re-decides only when a recent event is displaced.

Deployment Arm Inventory (MobileWorld, B=4)

Single-round arms sharing roster, budget, rendering, and greedy decoding:

Arm	Full (117)	Mem.-crit. (62)	Ctrl. (55)
Frozen + selector (CausalCache)	**36.8**	**30.6**	43.6
HGKV+selector (3-run mean)	33.9	28.0	40.6
frozen-taught selector	33.3	25.8	41.8
multi-slot-corpus-taught selector	33.3	24.2	43.6
$k \leq 1$-capped selector	28.2	19.4	38.2
OCR-text restoration	32.5	24.2	41.8
verbatim-text restoration	27.6	18.0	38.2
Recent-4 (3-run mean)	30.2	19.4	42.4

Table S10: All deployment arms at the primary budget. Task-paired bootstrap contrasts (20,000 replicates): CausalCache vs. Recent-4 full +6.6pp $[+0.3, +13.1]$ (p=0.035), memory-critical +11.3pp $[+2.7, +20.4]$ (p=0.006); CausalCache vs. HGKV+selector full +2.9pp $[-2.9, +8.6]$ (indistinguishable); $k \leq 1$ cap vs. HGKV+selector memory-critical -8.6pp $[-16.7, -1.6]$ (p=0.02); verbatim text vs. HGKV+selector memory-critical -10.4pp $[-18.6, -3.3]$ (p=0.006); OCR vs. Recent-4 memory-critical +4.8pp $[-2.2, +12.4]$.

Teacher Ablations

The deployed selector distills set utilities scored under the HGKV instrument. Set scoring covers 3,992 of the 5,335 eligible desktop states (58k singleton + 110k unique scored set edges); the remaining 1,343 states were never scored, so the deployed selector is trained on 75% of its intended label pool. Two retrained variants swap only the label teacher, with set labels rescored on a uniformly interleaved 60% state subset of the eligible pool (identical tree parameters), giving them 3,210 states and 88.8k set edges—80% of the deployed selector's realized volume, not the 60% the design intended relative to it. Scoring under the fully frozen policy yields 25.8% memory-critical—still significantly above Recent-4 (+6.5pp, p=0.04)—and scoring under an HGKV variant retrained on selector-composed multi-slot corpora yields 24.2%. The HGKV-taught deployed selector reaches 30.6%. At 80% relative volume the gap is therefore not primarily a volume artifact, though volume is not fully controlled either.

Teacher × deployment at 9% volume. To separate teacher identity from deployment pairing we ran a full factorial at a deliberately small matched volume (480 states, 9%; both teachers restricted to the same `dp_id` list, identical hyperparameters), each cell a full 117-task closed-loop round. Matching the selector to the policy that taught it—rather than pairing it with the frozen policy—is worth +1.7pp (CI $[-3.4, +6.8]$), and swapping the teacher at equal volume is worth -2.6pp (CI $[-9.4, +4.3]$); a same-selector cross-host replication returns 0.0pp (CI $[-6.8, +6.8]$), calibrating the noise floor. Neither factor is detectable. The informative outcome is the level rather than the contrasts: all four cells land at 28–31%, *below* frozen + Recent-4 (33.3%) and far below the full-volume selector (36.8%). At 9% of the labels a selector actively hurts, and no choice of teacher or deployment pairing recovers it. Label volume, not teacher identity, is the binding constraint in this regime—which is also why the 80%-volume teacher comparison above is the one that speaks to teacher quality.

Keeping the Probe at Deployment (2×2)

The probe is a training-time instrument, but nothing forces us to remove it at deployment. We therefore crossed policy side (frozen / gated last-36 / layer- and rank-matched ungated) with allocation (Recent-4 / selector) on the full 117-task roster, all cells run against the same emulator batch in a single round:

Policy side	Recent-4	+ selector
Frozen	33.3 / 25.8	36.8 / 30.6†
Gated last-36	31.6 / 29.0	35.9 / 30.6
Ungated full-layer	30.8 / 27.4	35.0 / 29.0

Table S11: Full roster / memory-critical success (%). †three-run mean from the main results, not rerun in this batch; the other five cells are same-batch.

Three readings. First, the selector is worth +4.3pp on *both* adapters (CI $[-1.7, +10.3]$ gated, $[-2.6, +11.1]$ ungated)—numerically identical, so the gain comes from reallocating memory and not from what the policy carries. Second, gating versus no gating is undetectable under either allocation (+0.9pp with Recent-4, likewise with the selector); the two adapters disagree on only 5 of 117 tasks. Third, no adapter cell beats frozen + selector: the best is 35.9% against 36.8%. Keeping the probe at deployment buys nothing, which is the empirical content of shipping a frozen policy.

Notably the offline instruments do not predict any of this. The ungated control has twice the gated adapter's offline selectivity (+0.0291 against +0.0145 on the B=4 corpus) and six times its wrong-history drift (0.031 against 0.005), yet the two are indistinguishable in closed loop on both allocation rules. Offline DiD separates instruments; it does not forecast deployment behavior.

Cross-Backbone Gates (GUI-Owl-1.5-32B)

The v4 recipe—same corpus, objective, 300 steps, checkpoint cadence 50—applied unchanged to the 32B backbone (world size 3, effective batch 6):

Checkpoint	DiD select	95% CI	A_r
step 50	+0.0009	$[-0.0002, +0.0020]$	+0.0005
step 100	+0.0067	$[+0.0048, +0.0088]$	−0.0023
step 150	+0.0049	$[+0.0031, +0.0068]$	+0.0014
step 200 (selected)	**+0.0142**	$[+0.0109, +0.0176]$	−0.0063
step 250	+0.0126	$[+0.0099, +0.0155]$	−0.0029
step 300	+0.0126	$[+0.0097, +0.0156]$	−0.0017

Table S12: 32B DiD gates on the 94 development groups (B=1). Five of six checkpoints pass all pre-specified gates; the frozen selection rule picks step 200. Recent-window drift stays within a third of the 0.02 cap throughout: token-gated selectivity inside the drift envelope transfers across backbone scale without retuning.

32B closed loop, and why the gates do not carry it. Mounting the 8B-taught selector on the frozen 32B policy scores 39.2% against 49.0% for 32B + Recent-4 (-9.8pp, paired $n{=}51$, CI $[-21.6, +2.0]$, 2 discordant pairs favouring the selector against 7 against). The interval crosses zero, but the sign held at all six interim samples (-5.3 / -12.5 / -10.9 / -11.8 / -10.2 / -9.8), so we read this as a real negative transfer rather than noise.

The mechanism is visible in the baseline: 32B + Recent-4 scores 49.0% where 8B + Recent-4 scores 30.2%. A backbone that recovers from the recent window most of what 8B needed archived pixels for has a different set of deficits, and an 8B probe measures 8B's. Reproducing the gain at 32B requires rescoring set utilities with the 32B policy in the probe.

Coverage. This is not a full-roster result. The Recent-4 arm reached $111/117$ and the selector arm $57/117$, intersecting on the 51 paired tasks reported above. The selector arm was paused partway to free capacity for the baseline and never resumed; the Recent-4 arm's last six tasks failed three attempts each with emulator-internal timeouts (container and HTTP service alive, screenshot never returning) and were abandoned as infrastructure failures after two emulator replacements. Any use of this comparison should carry the $n{=}51$ qualifier.